

Weekly Problem 7

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Exercise 1

Part (a)

1	$p \Rightarrow (p \Rightarrow p)_{\emptyset}$	
2	$p \Rightarrow q_{\{1\}}$	
3	$p_{\{2\}}$	
4	$p \Rightarrow (p \Rightarrow p)_{\emptyset}$	R1
5	$p \Rightarrow p_{\{2\}}$	$\rightarrow_R E4, 3$
6	$p_{\{3\}}$	
7	$p \Rightarrow p_{\{2\}}$	R5
8	$p_{\{2,3\}}$	$\rightarrow_R E7, 6$
9	$p \Rightarrow q_{\{1\}}$	R2
10	$q_{\{1,2,3\}}$	$\rightarrow_R E9, 8$
11	$p \Rightarrow q_{\{1,2\}}$	$\rightarrow_R I6-10$
12	$p \Rightarrow (p \Rightarrow q)_{\{1\}}$	$\rightarrow_R I3-11$
13	$(p \Rightarrow q) \Rightarrow (p \Rightarrow (p \Rightarrow q))_{\emptyset}$	$\rightarrow_R I2-12$

We then see that $p \Rightarrow (p \Rightarrow p) \vdash (p \Rightarrow q) \Rightarrow (p \Rightarrow (p \Rightarrow q))$ is a valid argument in relevant logic.

This derivation satisfies the entailment restriction since only formulas of the form $A \Rightarrow B_{\alpha}$ have been reiterated into subproofs.

Part (b)

1			$p \Rightarrow (q \Rightarrow p)_{\{1\}}$	
2			$q_{\{2\}}$	
3			$p_{\{3\}}$	
4			$p \Rightarrow (q \Rightarrow p)_{\{1\}}$	R1
5			$q \Rightarrow p_{\{1,3\}}$	$\rightarrow_R$ E4, 3
6			$q_{\{2\}}$	R2
7			$p_{\{1,2,3\}}$	$\rightarrow_R$ E5, 6
8			$p \Rightarrow p_{\{1,2\}}$	$\rightarrow_R$ I3-7
9			$q \Rightarrow (p \Rightarrow p)_{\{1\}}$	$\rightarrow_R$ I2-8
10			$(p \Rightarrow (q \Rightarrow p)) \Rightarrow (q \Rightarrow (p \Rightarrow p))_{\emptyset}$	$\rightarrow_R$ I1-9

We then see that $\vdash (p \Rightarrow (q \Rightarrow p)) \Rightarrow (q \Rightarrow (p \Rightarrow p))$ is a valid argument in relevant logic.

This derivation does not satisfy the entailment restriction since $q_{\{2\}}$ was reiterated into a subproof on line 6.